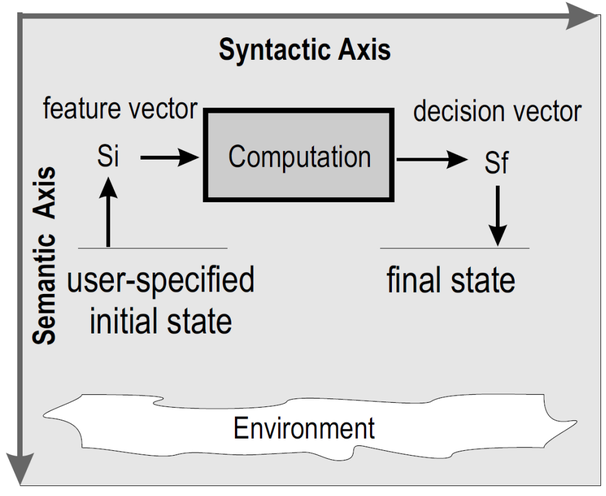
APRENDIZAGEM

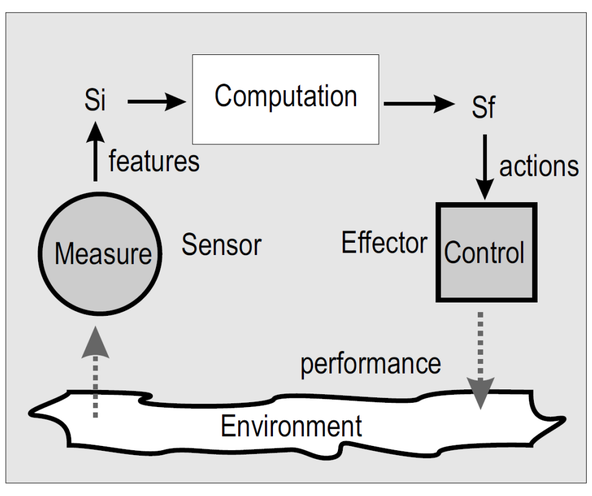
APRENDIZAGEM

## Cariani's robotic devices

(from Luiz E. Merkle,PhD Thesis, based on Cariani – 1991a,b,c)




(a) Non-adaptative computational device

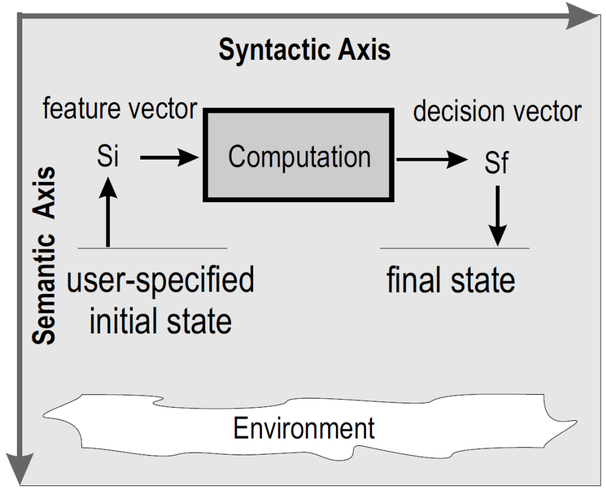



(b) Non-adaptative robotic device

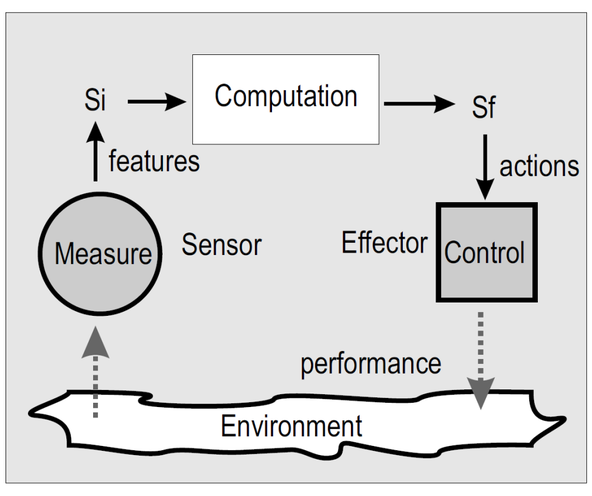
- Luiz Ernesto Merkle. Disciplinary and Semiotic Relations across Human–Computer Interaction. PhD Thesis, University Of Western Ontario, Canadá, 2002
- Peter Cariani. Adaptivity and emergence in organisms and devices. World Futures, 32:111{132, 1991a.
- Peter Cariani. Emergence and arti¯cial life. In C.G. Langton, C. Taylor, J.D. Farmer, and S. Ras mussen, editors, Arti¯cial Life II , SFI Studies in the Sciences of Complexity, volume X, pages 775{797. Addison–Wesley, 1991b.
- Peter Cariani. Some epistemological implications of devices which construct their own sensors and efectors. In Francisco F. Varela and P. Bourgine, editors, Proceedings of the First European Workshop on Arti¯cial Life, pages 484{493, Cambridge, MA, USA, 1991c.
- Peter Cariani. Cybernetic systems and the semiotics of translation. Athanor. Semiotica, Filosofia, Arte, Literature XI, 2, 200. Meltemi, Rome, 2001.

## Dispositivos robóticos de Cariani

(from Luiz E. Merkle,Tese de doutorado, baseada em Cariani - 1991a,b,c)




(a) Non-adaptative computational device




(b) Non-adaptative robotic device

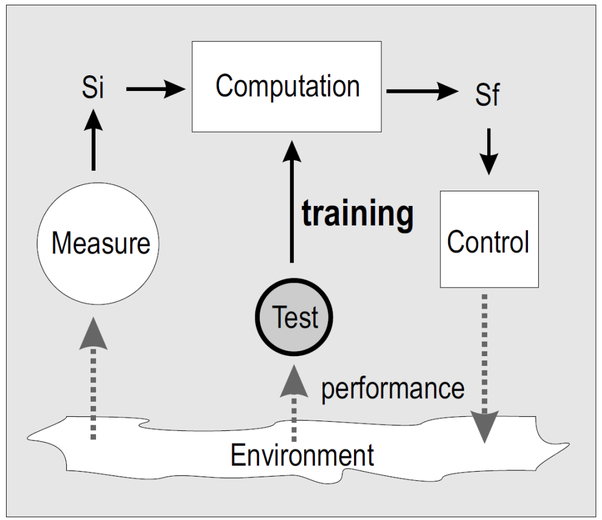
● Luiz Ernesto Merkle. Relações disciplinares e semióticas na interação humano-computador. Tese de Doutorado, University Of Western Ontario, Canadá, 200 ● Peter Cariani. Adaptabilidade e emergência em organismos e dispositivos. Futuros Mundiais, 32:111{132, 1991a. ● Pedro Cariani. Emergência e vida artificial. Em C.G. Langton, C. Taylor, J.D. Farmer e S. Ras mussen, editores, Arti¯cial Life II, SFI Studies in the Scienc é de

Complexidade, volume X, páginas 775{797. Addison-Wesley, 1991b. ● Pedro Cariani. Algumas implicações epistemológicas de dispositivos que constroem seus próprios sensores e efetores. Em Francisco F. Varela e P. Bourgine, edi tores,

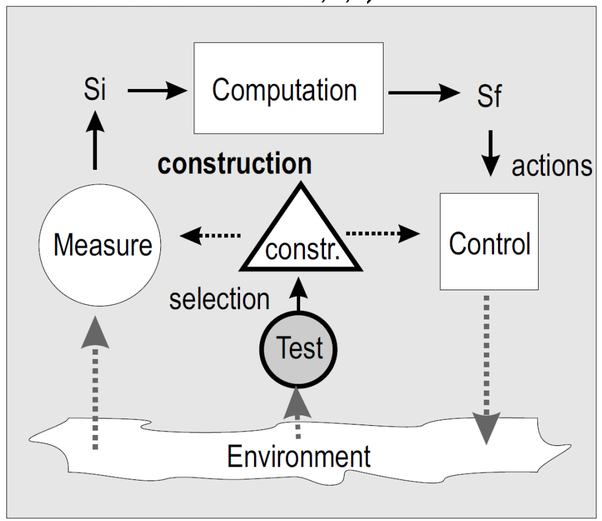
Anais do Primeiro Workshop Europeu sobre Vida Artificial, páginas 484{493, Cambridge, MA, EUA, 1991c. ● Pedro Cariani. Sistemas cibernéticos e semiótica da tradução. Atanor. Semiótica, Filosofia, Arte, Literatura XI, 2, 200. Meltemi, Roma, 2001.

# Cariani's robotic devices

(from Luiz E. Merkle,PhD Thesis, based on Cariani – 1991a,b,c)




(c) Syntactically adaptive robotic device

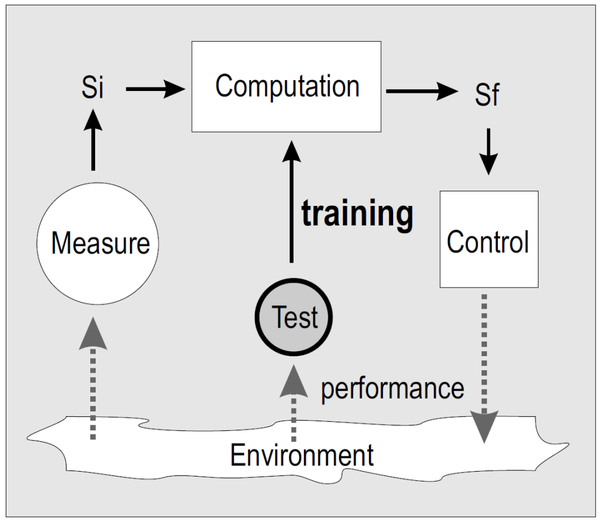



(d) Semantically adaptive robotic device

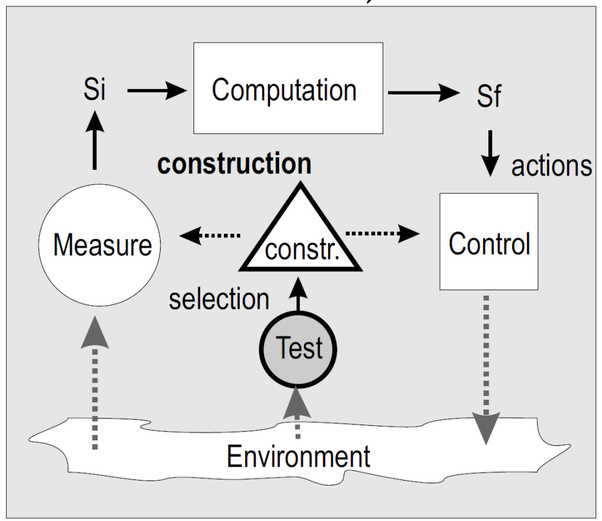
- Luiz Ernesto Merkle. Disciplinary and Semiotic Relations across Human–Computer Interaction. PhD Thesis, University Of Western Ontario, Canadá, 2002
- Peter Cariani. Adaptivity and emergence in organisms and devices. World Futures, 32:111{132, 1991a.
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# Dispositivos robóticos de Cariani

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(c) Syntactically adaptive robotic device




(d) Semantically adaptive robotic device

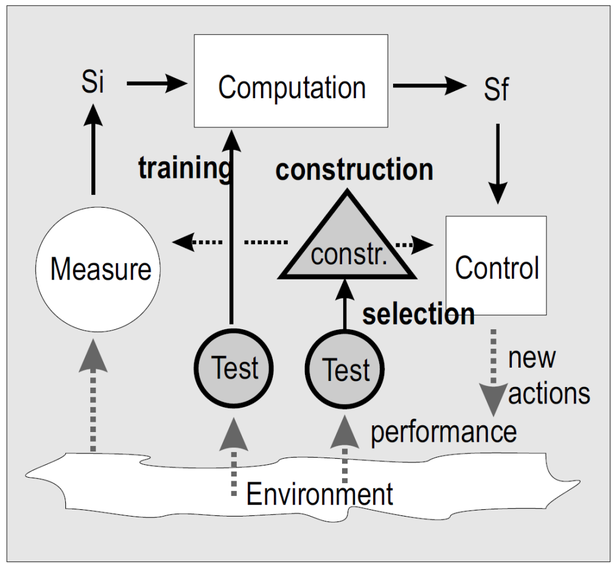
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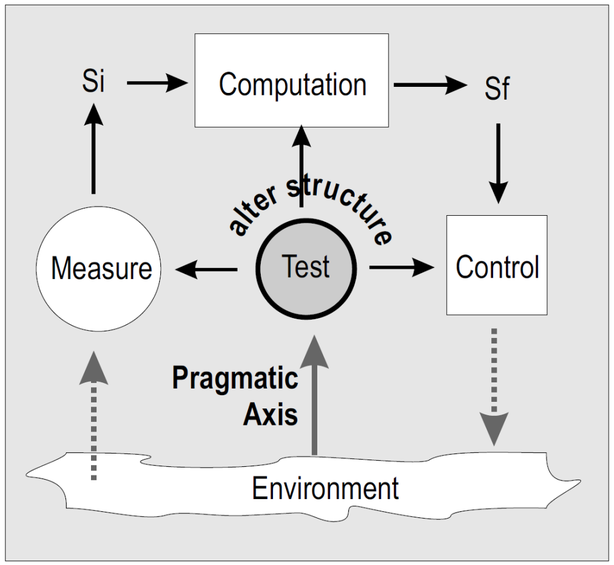
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# Cariani's robotic devices

(from Luiz E. Merkle,PhD Thesis, based on Cariani – 1991a,b,c)




**(e) Semantically and syntactically adaptive robotic device**

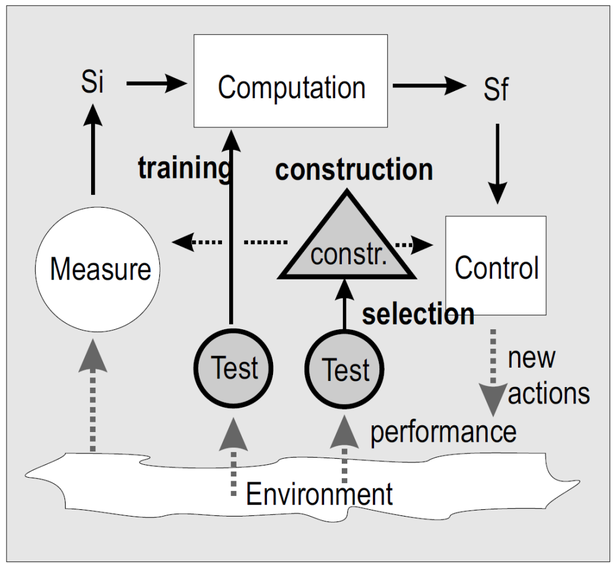



**(f) Evolutionary device**

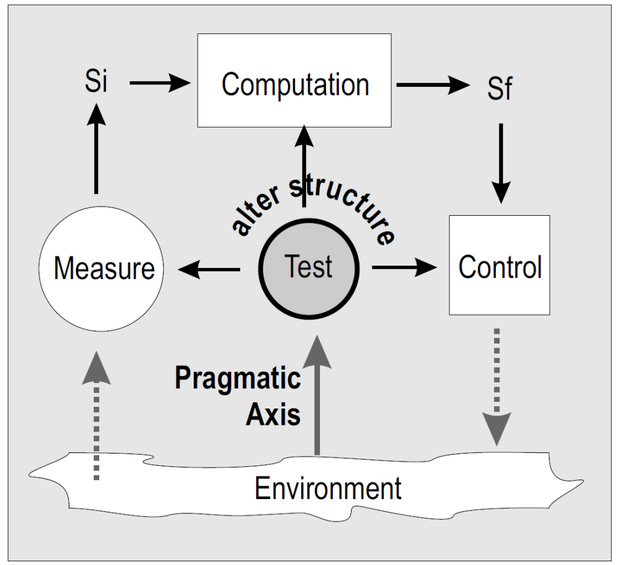

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Dispositivos robóticos de Cariani

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**(e) Semantically and syntactically adaptive robotic device**




**(f) Evolutionary device**


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